FYDP I · UNITED INTERNATIONAL UNIVERSITY · DEPT. OF CSE

Solving the Lorenz ODE System using Optimal ANN Architectures

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Introduction & Motivation

Coupled nonlinear ODEs like the **Lorenz-1960** system — an early meteorological model — rarely have closed-form solutions. Classical solvers (Runge–Kutta) are accurate but **mesh-dependent** and re-solve step-by-step for every query. A trained network can instead learn the continuous map **t → (x, y, z)**, yet most neural ODE work relies on physics-informed losses or operator learning.

Central question: can a **plain feedforward ANN** — trained only on numerical data, no physics in the loss — approximate the Lorenz-1960 solution, and **which architecture does it best?**

Objectives

- Generate a trusted reference with RK4 **and** SciPy; cross-validate the two.
- Systematically vary **depth**, **width**, and **activation** (plain ANN).
- Measure accuracy vs cost: MSE, MAE, max error, training / inference time.
- Benchmark the best plain ANN against PINN / DeepONet results.

Research Questions & Gap

Gap. No prior work runs a systematic plain-ANN architecture study on Lorenz-1960; existing studies fix the architecture, use only simple ODEs, or rely on physics-informed losses.

- Can a plain ANN learn the solution from numerical data alone?
- Which depth × width × activation gives the best accuracy–cost trade-off?
- How close does it come to physics-informed baselines?

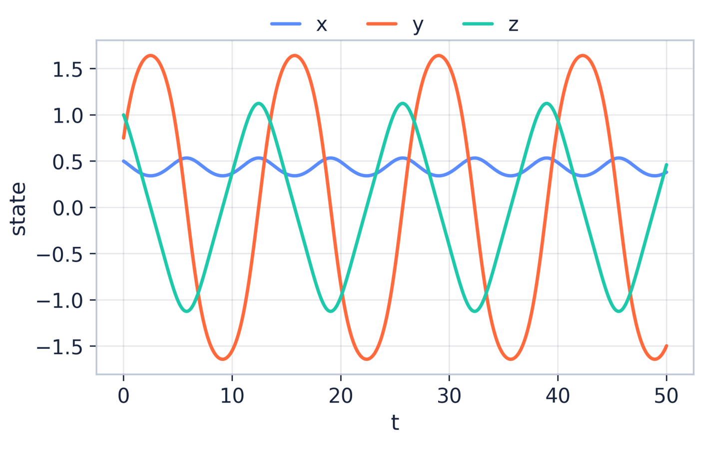
Background: The System

$$\dot{x} = -0.10\ y\ z$$

$$\dot{y} = +1.60\ x\ z$$

$$\dot{z} = -0.75\ x\ y$$

$k=2,\ l=1 \cdot x_0=0.5,\ y_0=0.75,\ z_0=1 \cdot t\in[0,1]$



Bounded, quasi-periodic orbits, t∈[0,50].

Complex Engineering Problem

Mapped to the CEP rubric — **problem solving** (depth of knowledge, conflicting requirements, depth of analysis, interdependence), **activities** (resources, interaction, innovation, unfamiliarity), and **knowledge** (natural sciences, mathematics, engineering fundamentals, specialist ML, research literature, ethics).

Methodology: Two-Phase Study

Phase 1 — Architecture search

depth {1, 2, 3, 4} × width {20, 50, 100} × activation {tanh, ReLU, sigmoid, GELU, Swish}

= 60 runs · Adam fixed

Phase 2 — Optimizer comparison

3 best architectures × {Adam, L-BFGS, SGD + momentum}, everything else constant.

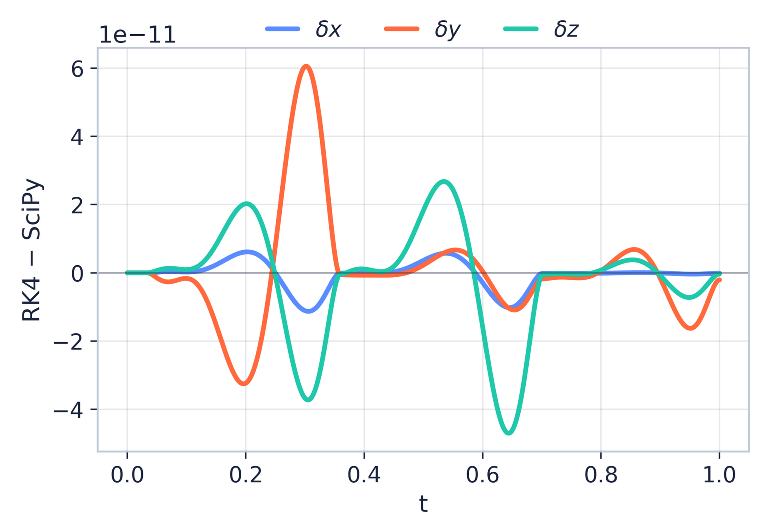
= 9 runs

69 controlled experiments — splitting the phases cleanly separates architecture effects from optimizer effects.

Generate RK4 + SciPy → Preprocess 80/20 split → Train ANN grid → Evaluate 5 metrics → Select best

Numerical Baseline & Validation (Result)

The reference trajectory is generated by two independent solvers and cross-checked before any ANN sees it: a custom **RK4** integrator ($h = 10^{-3}$, 1001 points) and **SciPy DOP853** ($rtol = 10^{-10}$), with an 80/20 train–validation split (seed 42).



RK4 - SciPy disagreement stays at the 10^{-11} level.

1.8×10^{-22}

solver-agreement MSE (RK4 vs SciPy)

✓ target was $< 10^{-10}$

A defensible **ground truth** for all FYDP-II ANN training.

Conclusion & Future Work

Status (FYDP-I): research question framed, 17-paper review with an explicit gap, two-phase design (69 runs) locked, dual-solver baseline verified. No accuracy claims are made until the FYDP-II experiments produce evidence.

- Build the ANN training pipeline; run Phase 1 (60), then Phase 2 (9).
- Repeated seeds for stability; denser grids and new initial conditions.
- Direct comparison with PINN / operator-learning baselines.

Selected References

[1] Lorenz (1960), *Tellus* 12(3).
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[5] Lau & Werth (2020), ODEN.
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[7] Chen et al. (2018), Neural ODEs, *NeurIPS*.
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